

PROJECT

Deadline: June 2nd 2026

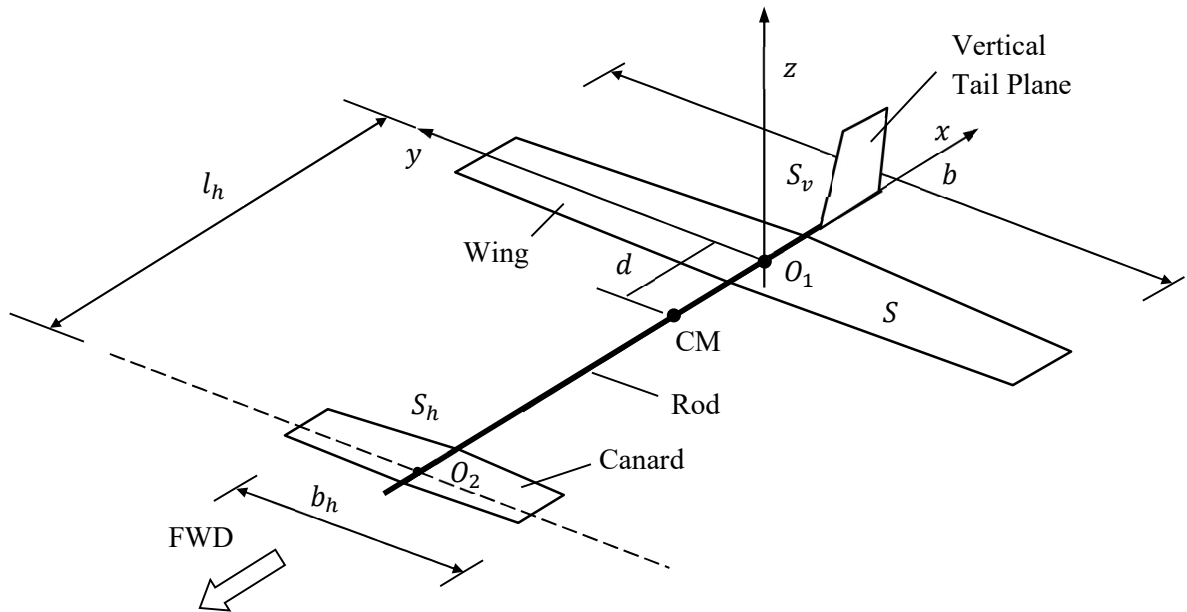
Description

Consider a glider of canard configuration and, for a very preliminary simplified analysis, modeled as a wing attached to a canard (horizontal forward surface) and a vertical tail plane (VTP) by means of a thin rod, this last with negligible aerodynamic effect, as sketched in the picture below. Both the wing and canard are straight (zero sweep) and of trapezoidal planform, with their respective wingspans and chords at root and tip being

$$b = 15 \text{ m} \quad c_r = 0.95 \text{ m} \quad c_t = 0.55 \text{ m}$$

$$b_h = 3.0 \text{ m} \quad c_{rh} = 0.5 \text{ m} \quad c_{th} = 0.3 \text{ m}$$

The horizontal distance between points O_1 and O_2 is $l_h = 4.0 \text{ m}$ and $l_v = 1.2 \text{ m}$, and the area of the vertical tail plane is $S_v = 1.5 \text{ m}^2$.



The wing is fitted with a modified HQ 300 airfoil, and the chord of the center section of the wing is assumed to have zero incidence angle $i_w = 0$ with respect to the rod.

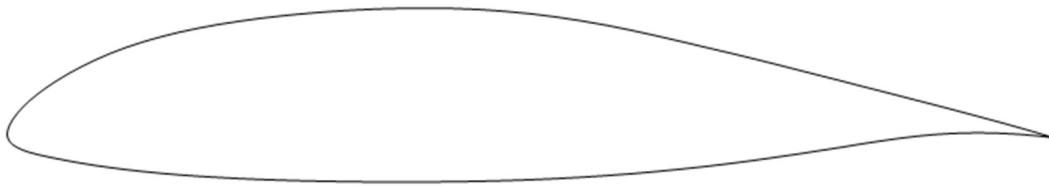
The canard is fitted with a two-element system consisting of NACA 0012 airfoils, and its incidence angle with respect to the rod is $i_h = 3^\circ$.

PART 1: THE CONSTANT STRENGTH VORTEX METHOD APPLIED TO SIMPLE AND TWO-ELEMENT AIRFOILS

This part focuses on the two-dimensional incompressible potential flow around a HQ 300 airfoil and a system consisting of two NACA 0012 airfoils, their geometries being given by the coordinates of discrete points listed in the txt files provided (for ever increasing number of divisions in powers of 2).

For all the questions below, the results obtained must be adequately commented and justified.

Consider initially the HQ 300 airfoil.



The following is requested:

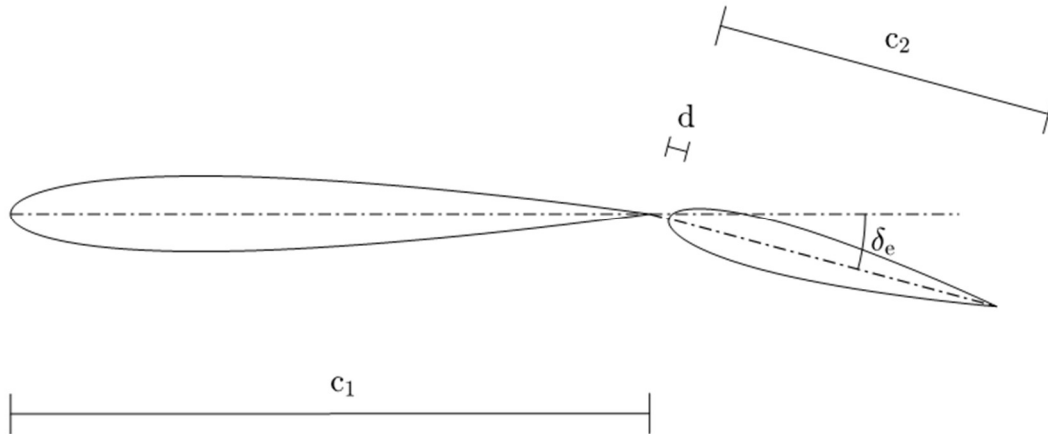
1. Obtain the lift coefficient C_l and the pitching moment coefficient about the quarter chord point $C_{m_{1/4}}$ for the angles of attack varying in a range $0 \leq \alpha \leq 8^\circ$.

HQ 300	$\alpha = 0^\circ$	$\alpha = 2^\circ$	$\alpha = 4^\circ$	$\alpha = 6^\circ$	$\alpha = 8^\circ$
C_l					
$C_{m_{1/4}}$					

Determine also the position of the aerodynamic center.

2. Using Kármán-Tsien's rule, determine the critical Mach number for $\alpha = 0^\circ, 2^\circ, 4^\circ$.
3. Compute the lift coefficient for $\alpha = 4^\circ$, for the following free stream Mach numbers:
 $M_\infty = M_{cr} - 0.15$, $M_\infty = M_{cr} - 0.10$, $M_\infty = M_{cr} - 0.05$, $M_\infty = M_{cr}$.

Consider now a system consisting of two symmetric NACA 0012 airfoils arranged in tandem as depicted in the figure below, with their respective chords $c_1 = 0.64$ and $c_2 = 0.34$, and gap $d = 0.02$. The angle δ_e can vary both up and down.



The following is requested:

- Obtain the lift coefficient C_l , and the pitching moment coefficient about the quarter chord point $C_{m_{1/4}}$, for zero elevator deflection, and the angle of attack varying in a range $0 \leq \alpha \leq 8^\circ$.

Compound Airfoil	$\alpha = 0^\circ$	$\alpha = 2^\circ$	$\alpha = 4^\circ$	$\alpha = 6^\circ$	$\alpha = 8^\circ$
C_l					
$C_{m_{1/4}}$					

- Compute the lift coefficient and the moment coefficient about the original quarter chord point, for an angle of attack $\alpha = 4^\circ$, and the elevator deflection angle varying in a range $0 \leq \delta_e \leq 16^\circ$.

Compound Airfoil	$\delta_e = 0^\circ$	$\delta_e = 4^\circ$	$\delta_e = 8^\circ$	$\delta_e = 12^\circ$	$\delta_e = 16^\circ$
C_l					
$C_{m_{1/4}}$					

PART 2: PRANDTL'S LIFTING LINE MODEL APPLIED TO COMPOUND WINGS OF LARGE ASPECT RATIO

This part focuses on the application of Prandtl's Lifting Line Model for the study of the aerodynamics of single and compound wings of large aspect ratio, using the section (two-dimensional) lift and moment coefficients determined in part 1.

For all the questions below, the results obtained must be adequately commented and justified.

Where applicable, the following expressions for the viscous drag coefficient will be assumed:

$$C_d = 0.0183 C_l^2 - 0.0302 C_l + 0.0187 \quad \text{for HQ 300}$$

$$C_d = 0.0052 C_l^2 + 0.0071 \quad \text{for the canard}$$

$$C_d = 0.0062 \quad \text{for the VTP (because of symmetric flow)}$$

Study of the wing isolated

Considering initially the wing isolated, for an angle of attack of 4° (measured at the wing central section), the following is requested:

1. Select and justify an adequate wing twist, and plot the spanwise distribution of the local coefficients of lift, viscous drag, induced drag and induced angle of attack.

Study of the complete system

Considering now the complete wing, canard and VTP system, using the wing twist found in the previous question, the following is requested:

2. For an angle of attack of 4° (measured at the wing central section), and zero elevator deflection, plot the spanwise distribution of aerodynamic coefficients, and for both the wing and canard, and determine the required location of the center of mass CM to attain balance of pitching moments.
3. For a range of angles of attack $0^\circ \leq \alpha \leq 6^\circ$, and zero elevator deflection, plot the aerodynamic polar curve (C_L vs. C_D).
4. Compute the global lift coefficient and pitching moment coefficient about the CM when $\alpha = 4^\circ$ the elevator deflection is $\delta_e = 12^\circ$.